Benkang Lou

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Education Background

Southern University of Science and Technology (SUSTech) *Sep. 2022 – Jun. 2025*
M.Eng. in Intelligent Manufacturing and Robotics
Supervisor: Prof. Juan Yi and Prof. Jian S. Dai
Rank: 10/56 GPA: 3.65/ 4

University of Science and Technology Beijing (USTB) *Sep. 2018 – Jun. 2022*
B.Eng. in Mechanical Engineering (Honors)
Rank: 22/186 GPA: 3.54/ 4 (85.4/100)

Professional Experience

Southern University of Science and Technology (SUSTech) *Aug. 2025 – Present*
Research Assistant
Supervisor: Prof. Jian S. Dai

Publications

1. **Benkang Lou**, Juan Yi, Yang Yang, Sicong Liu, Jian S. Dai, Zheng Wang, "Soft Robotic Proprioception and Mutual Positioning via Magnetic Multifield Induction Sensing Array," *IEEE/ASME Transactions on Mechatronics*, 2026.
2. Juan Yi, Jiahao Xu, Chaoyi Huang, **Benkang Lou**, et al., "Actuator Heterogeneous Deformation with Optical Sensing for Proprioception and Exteroception of Soft Robots," *Sensors and Actuators: A.Physical*, 2026.
3. Juan Yi, Jiahao Xu, Yuxuan Deng, Yifan Xuan, Chaoyi Huang, Zhonggui Fang, **Benkang Lou**, et al., "Far-field Magnetic Sensing on Soft Origami Actuator for Spatial Multi-dimensional Movement and Force Perception," *Soft Robotics*, 2025.
4. **Benkang Lou**, Yifan Xuan, Lin Wang, Zhongfeng Qian, Rongwei Wen, Sicong Liu, Juan Yi, Jian S. Dai, Zheng Wang, "Mechanoreception of Soft Robots via Multi-channel Optical Sensing Using Ambient Light and Internal Infrared Light," *IEEE International Conference on Robotics and Biomimetics*, 2024.
5. Shaowu Tang, Kailuan Tang, Yutao Guo, Shijian Wu, Jiahao Xu, **Benkang Lou**, et al., "Soft Robotic Proprioception Enhancement via 3D-Printed Differential Optical Waveguide Design," *IEEE Robotics and Automation Letters*, 2024.

Works in Preparation

1. Xinyu Xiao, **Benkang Lou**, Jian S. Dai, Huijuan Feng, "Reconfigurable Mechanism Analysis of Hexaflexagon and Its Application in Robotic Structure Design."

Research Experience

An Underwater Metamorphic Quadruped Robot *Sep. 2025 – Mar. 2026*

- In progress

Crawling Origami Robot Inspired by Möbius Structure *Jun. 2024 – Mar. 2025*

- Design a novel crawling robot based on the Möbius strip, which features an continuous flipping capability.

- Spearheaded the robot prototype development, responsible for hardware system design and control strategies.

Magnetic Multifield Induction Sensing Array *Feb. 2023 – Nov. 2025*

- Proposed a multi-frequency magnetic multifield induction sensing method to distinguish spatial magnetic fields through frequency diversity. This method significantly extends the magnetic sensing range for soft robots.
- Designed soft robotic bending modules in which distinct magnetic field frequencies are allocated to individual modules, enabling modules to be stacked without magnetic interference.
- Magnetic field information between different modules can be mutually detected. Based on this capability, two planar crawling robots were designed, each able to sense the relative position of the other.

Multi-channel Optical Waveguide Sensor *Feb. 2023 – Mar. 2023*

- Proposed a multi-channel soft optical waveguide sensor using both ambient and infrared light.
- Designed a soft gripper integrated with the optical waveguide sensor, capable of sensing contact force and contact location simultaneously.

Heterogeneous Deformation with Optical Sensing of Actuator *Jul. 2022 – Jan. 2023*

- Participated in studying the deformation characteristics of origami-inspired actuators and sensing deformation using flexible optical waveguides.
- Developed and integrated optical waveguide sensor hardware into soft robotic platforms.
- Developed an untethered crawling robot with integrated locomotion and sensing capabilities, enabling remote-controlled crawling and trajectory perception.

Honors and Awards

Third Prize, 1st Continuum Manipulator Competition, SUSTech *2023*
Outstanding Graduate, USTB *2022*
First Prize, North China Region, 16th National College Student Smartcar Competition, USTB *2021*
First Prize, Beijing, 9th Beijing Undergraduate Engineering Training Competition, USTB *2021*
National Encouragement Scholarship (Two times), USTB *2020,2021*
National First Prize, 15th National College Student Smartcar Competition, USTB *2020*
Champion, North China Region, 15th National College Student Smartcar Competition, USTB *2020*
Renmin Third Scholarship, USTB *2019*

Technical Skills

Software: Keil, Arduino, MATLAB, SolidWorks, AutoCAD
Programming: C, Python, MATLAB
Language: Mandarin, English

Referees

Jian S. Dai *(jian.dai@kcl.ac.uk)*
IEEE Fellow, ASME Fellow
Fellow of the Royal Academy of Engineering, Member of Academia Europaea
Robotica Editor-in-Chief, Dean of SUSTech Institute of Robotics
Chair Professor, SUSTech and KCL